

Fractal Research Group / SoilPulse-ET

SmallSat onboard triage

water-stress follow-up

A compact satellite cannot downlink everything.

SoilPulse-ET decides what to sense, summarize, downlink, or defer.

What We Have Done

Use case

Water-stress triage for regenerative agriculture and land-resilience users.

NASA anchor

ECOSTRESS evapotranspiration data grounds the first data path.

Demo

A reproducible policy chooses priority chip, feature summary, or defer.

Budgets

The policy is constrained by downlink, packet count, processing time, and energy.

How It Works

NASA data -> tile state -> onboard packet choice

A tile is one candidate land area. Each tile gets a priority score and an evidence-quality score.

priority chip

Send a detailed observation when the tile looks urgent.

feature summary

Send compact features when a full chip is not needed.

defer

Wait when the evidence is cloudy, stale, or low-confidence.

What We Optimized

The 71.43% is byte savings, not the percent of tiles deferred.

Baseline

8 full chips

full chip for every candidate

Selected

2 chips + 2 summaries

four packets, not four full chips

Deferred

4 tiles

zero KiB this contact

Baseline size

784 KiB

all eight as full chips

Used size

224 KiB

96 + 112 + 8 + 8 KiB

Saved

560 KiB / 71.43%

bytes avoided, not tile count

Validation + Mission Hardening

Real data

Expand ECOSTRESS windows and replace synthetic support fixtures with real HLS and terrain ingestion.

Calibration

Tune chip / summary / defer thresholds against irrigation and conservation user priorities.

Generalization

Test across seasons, regions, water-stress windows, and land-cover types.

Hardware

Map the policy to a specific 6U compute, power, thermal, storage, and downlink stack.

Why This Matters

- More selective SmallSats can return higher-value observations.
- Onboard processing can reduce wasted downlink during constrained contacts.
- Land managers can get faster signals during water-stress windows.

SoilPulse-ET helps missions manage scarce satellite resources so land managers can respond faster to scarce water.